Baseball Statcast
2025 Out-of-Sample Forecast Report

Training period: 2019–2024 | **Validation period:** 2025

Models were trained exclusively on 2019–2024 data, then applied to 2025 players who were never seen during training. This report measures true out-of-sample accuracy using MAE, RMSE, R^2, and Percent Error — a more rigorous test than cross-validated R^2 on historical data.

Part 1 — Training Model Summary (2019–2024)

Models were selected and tuned using 2019–2024 data only. The best model per target (by held-out test R^2) was retained for 2025 deployment.

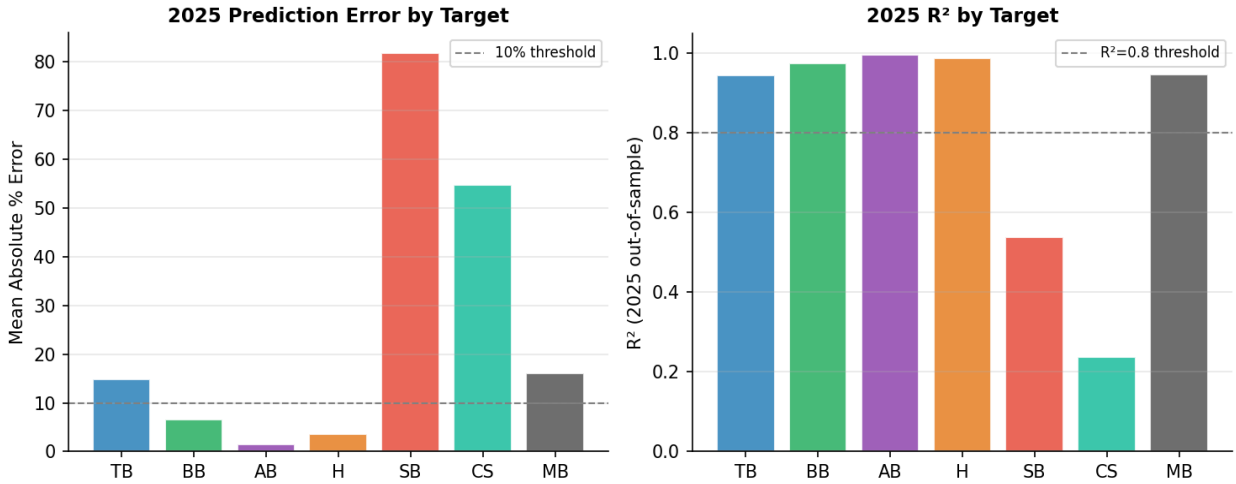
Target	Best Model	Train R^2	CV R^2	Train RMSE
TB	Linear Regression	0.9541	0.9484±0.0126	17.1072
BB	Lasso Regression	0.8929	0.8953±0.0149	7.6481
AB	Lasso Regression	0.9864	0.9888±0.0013	19.2832
H	Lasso Regression	0.9569	0.9587±0.0067	9.9207
SB	Poisson GLM (5 features)	0.4099	0.405	7.95
CS	Neg. Binomial GLM (5 features)	0.2863	0.2804	2.2468

Part 2 — 2025 Out-of-Sample Validation

The trained models were applied to 2025 batters. The table below shows how well the predictions match the actual 2025 statistics. R^2 here is *not* a training metric — it reflects genuine predictive accuracy on unseen data.

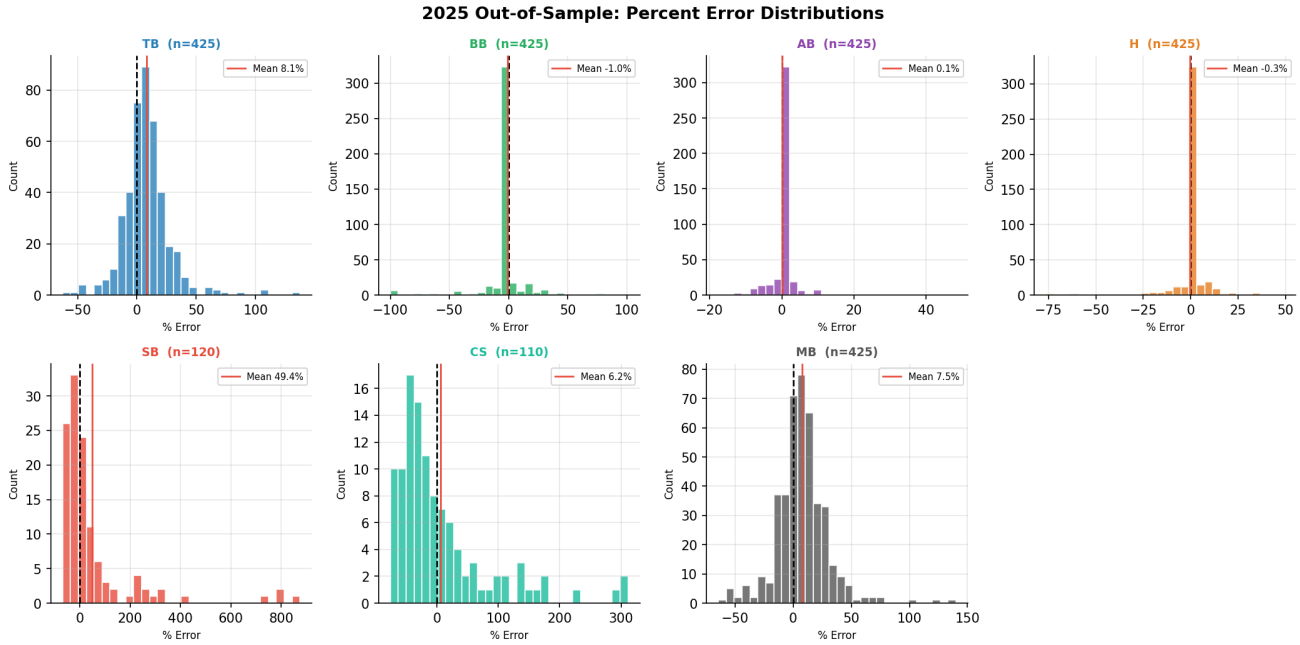
Target	n	R^2 (2025)	MAE	RMSE	Mean APE%	±5%	±10%	±20%
TB	425	0.9436	15.224	19.597	14.8%	24.9%	46.8%	76.9%
BB	425	0.9737	1.425	3.692	6.7%	78.1%	81.2%	90.1%
AB	425	0.9961	4.301	10.508	1.5%	88.9%	96.5%	99.1%
H	425	0.9868	2.314	5.372	3.6%	79.8%	88.7%	95.3%
SB	122	0.5384	5.174	7.219	81.9%	9.8%	16.4%	30.3%
CS	122	0.2369	1.854	2.421	54.8%	3.3%	12.3%	21.3%
MB	425	0.9449	5.238	6.953	16.1%	22.4%	43.8%	74.4%

Summary Overview



Left: Mean Absolute Percent Error per target. Lower is better. Right: Out-of-sample R^2 per target. Higher is better.

Percent Error Distributions

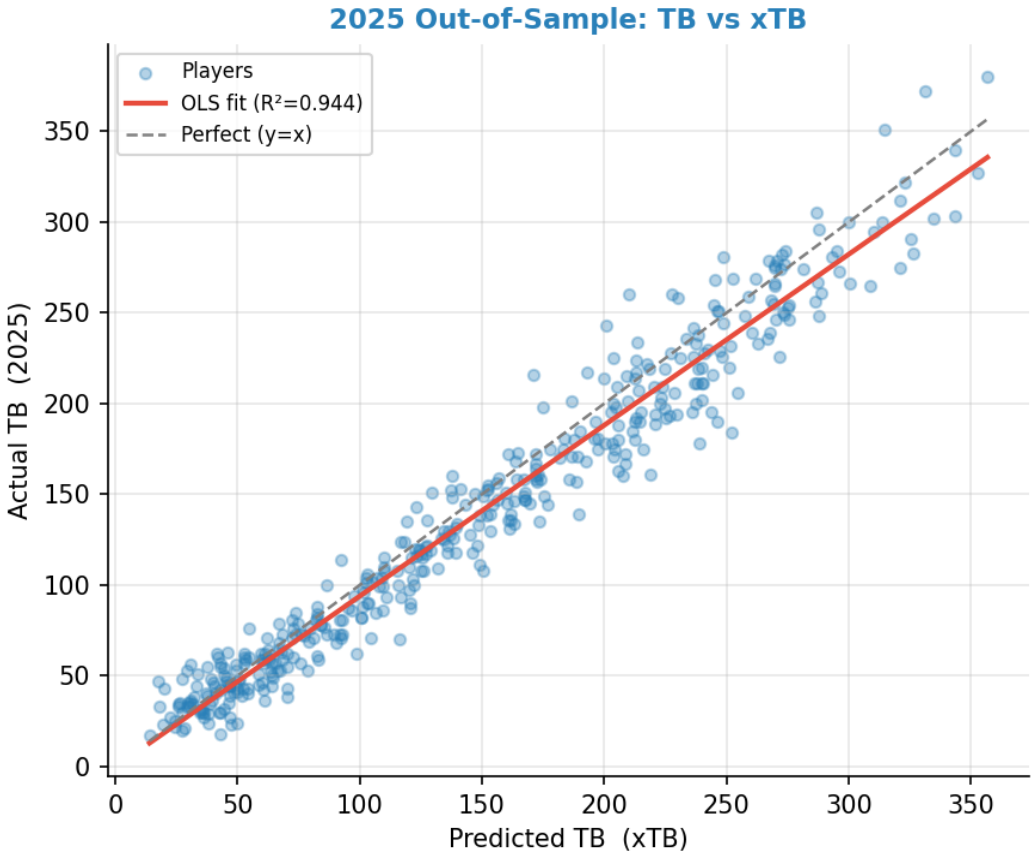


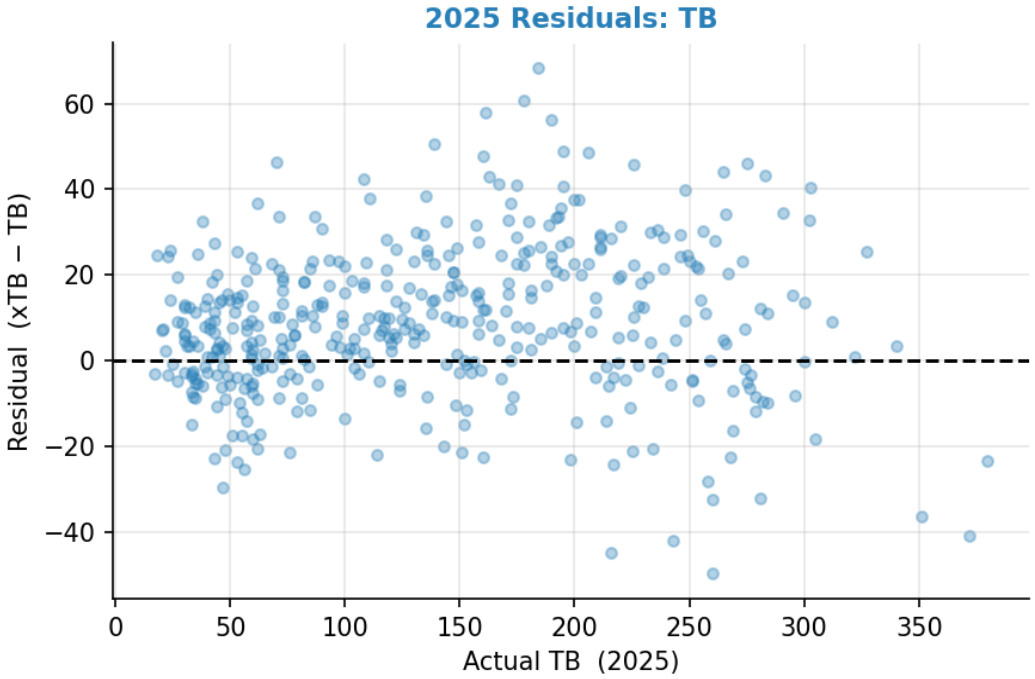
Percent error distributions for each target in 2025. A distribution centred on zero indicates unbiased predictions.

Per-Target: Predicted vs Actual (2025)

Target: TB

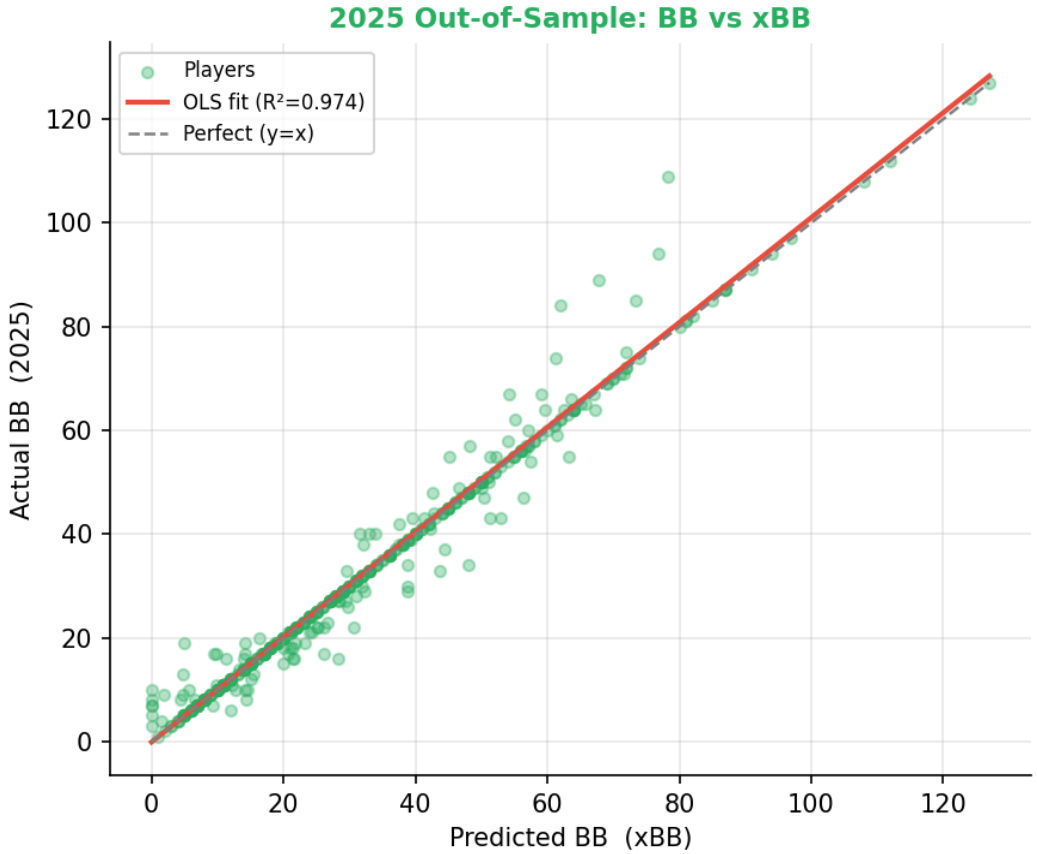
n=425 | R²=0.9436 | MAE=15.224 | RMSE=19.597 | Mean APE=14.76%

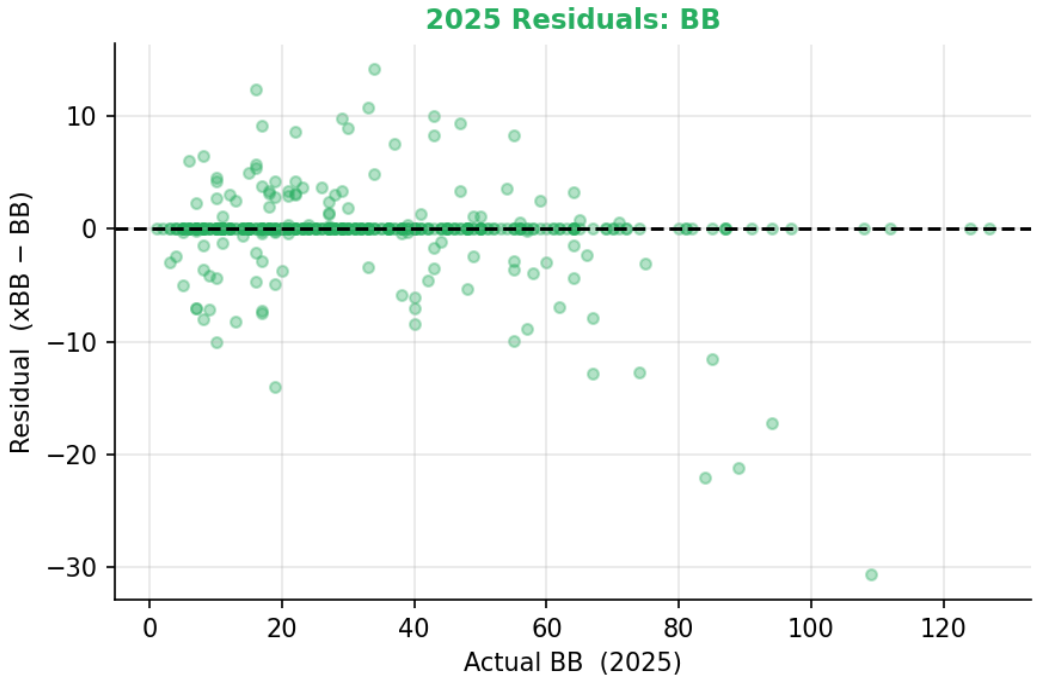




Target: BB

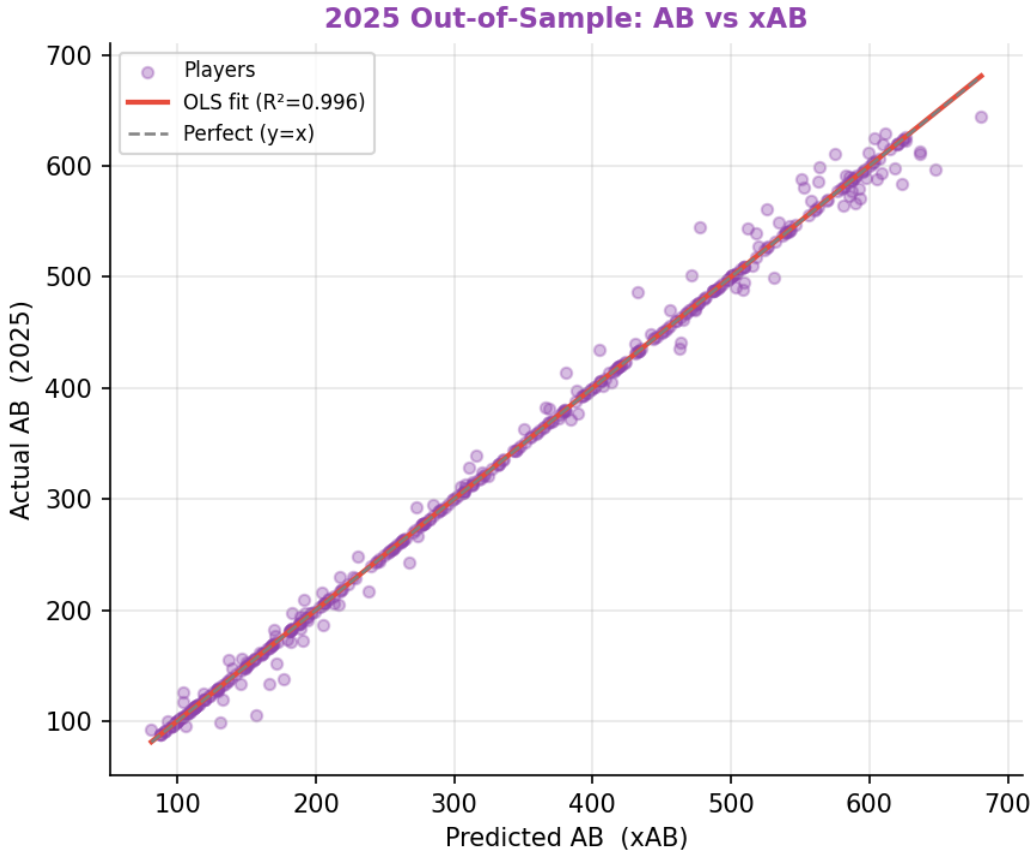
n=425 | R²=0.9737 | MAE=1.425 | RMSE=3.692 | Mean APE=6.65%

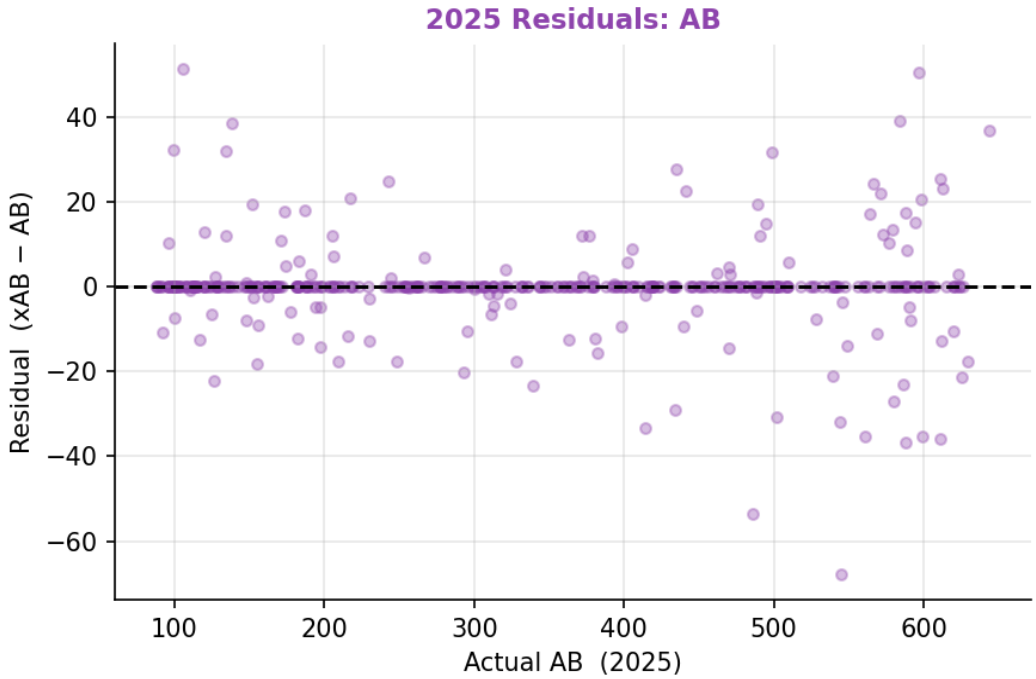




Target: AB

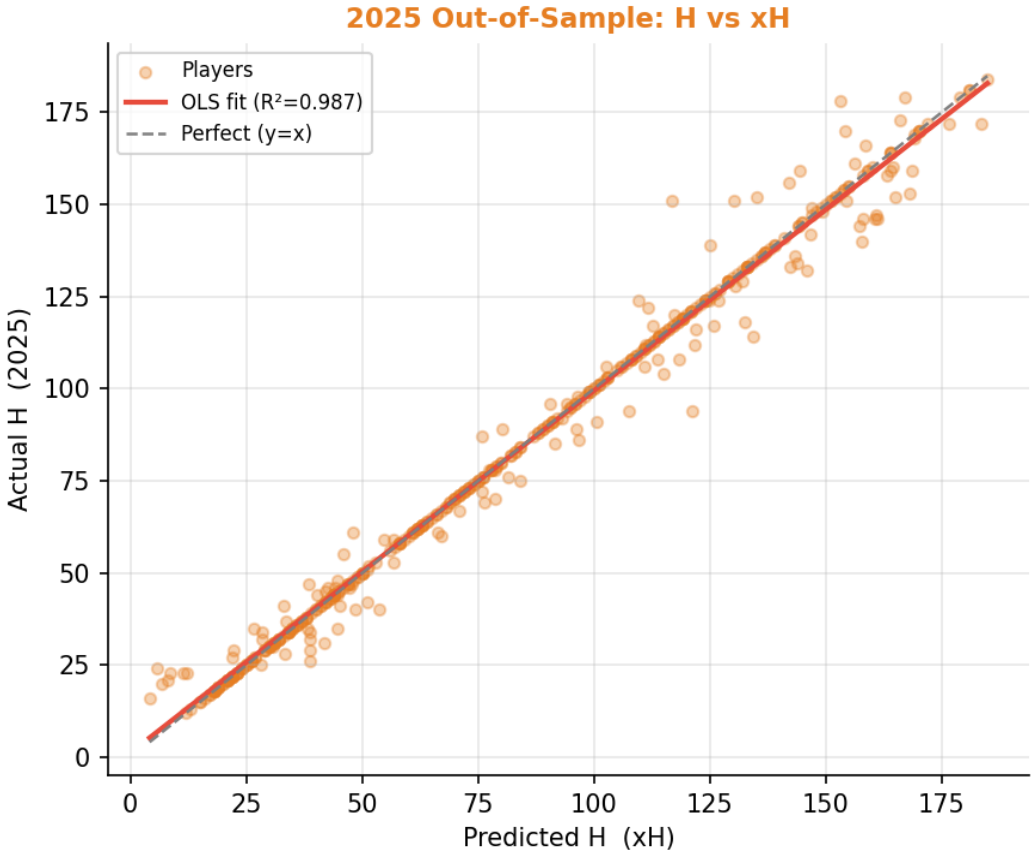
n=425 | R²=0.9961 | MAE=4.301 | RMSE=10.508 | Mean APE=1.48%

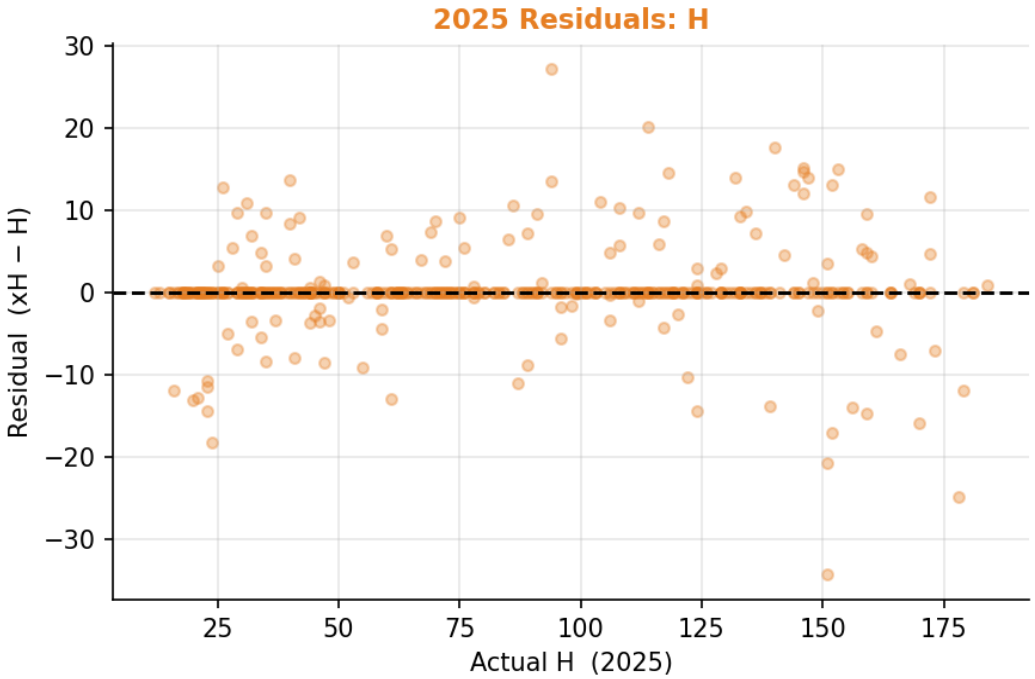




Target: H

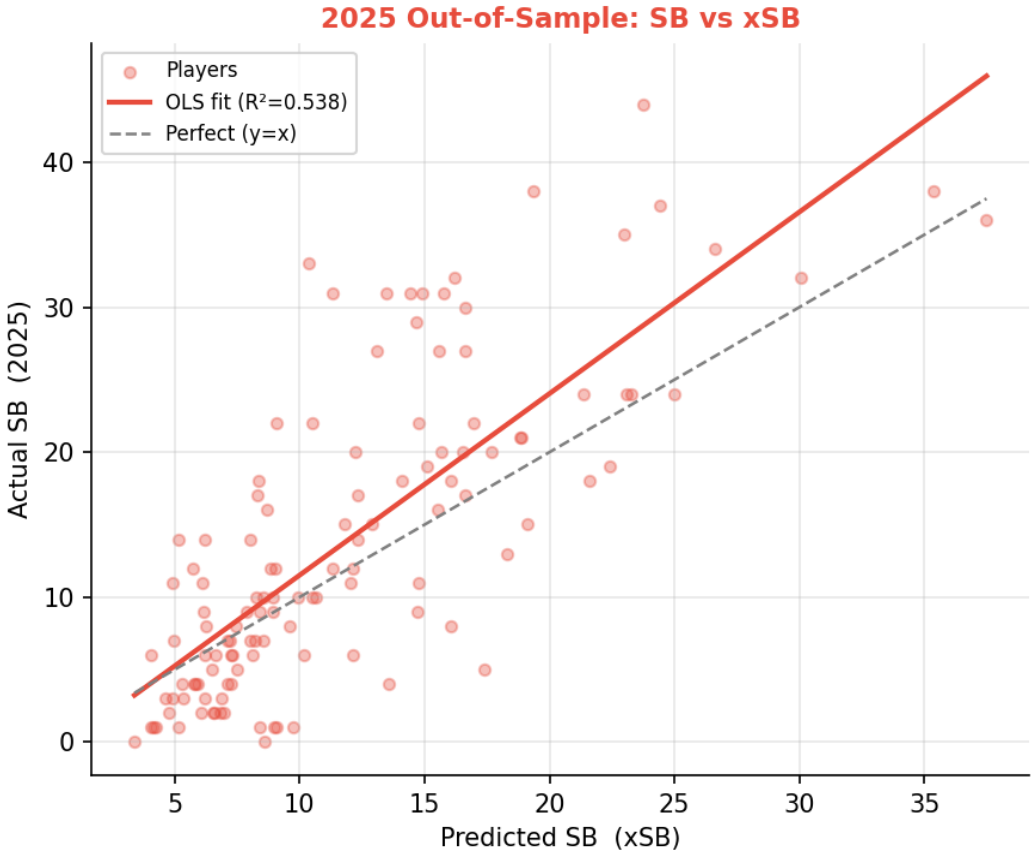
n=425 | R²=0.9868 | MAE=2.314 | RMSE=5.372 | Mean APE=3.65%

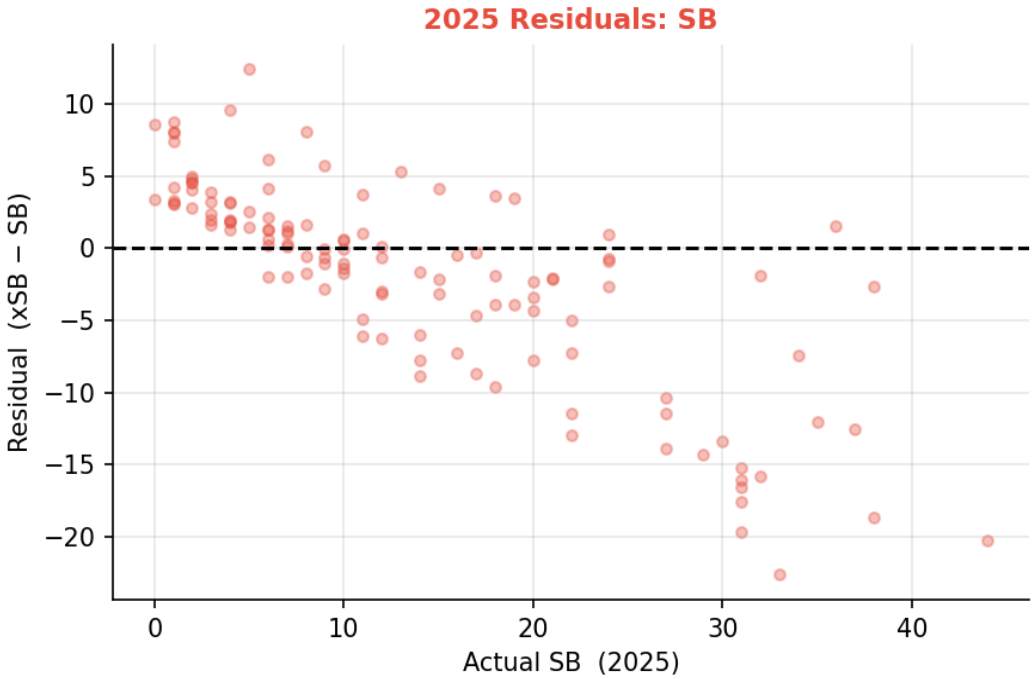




Target: SB

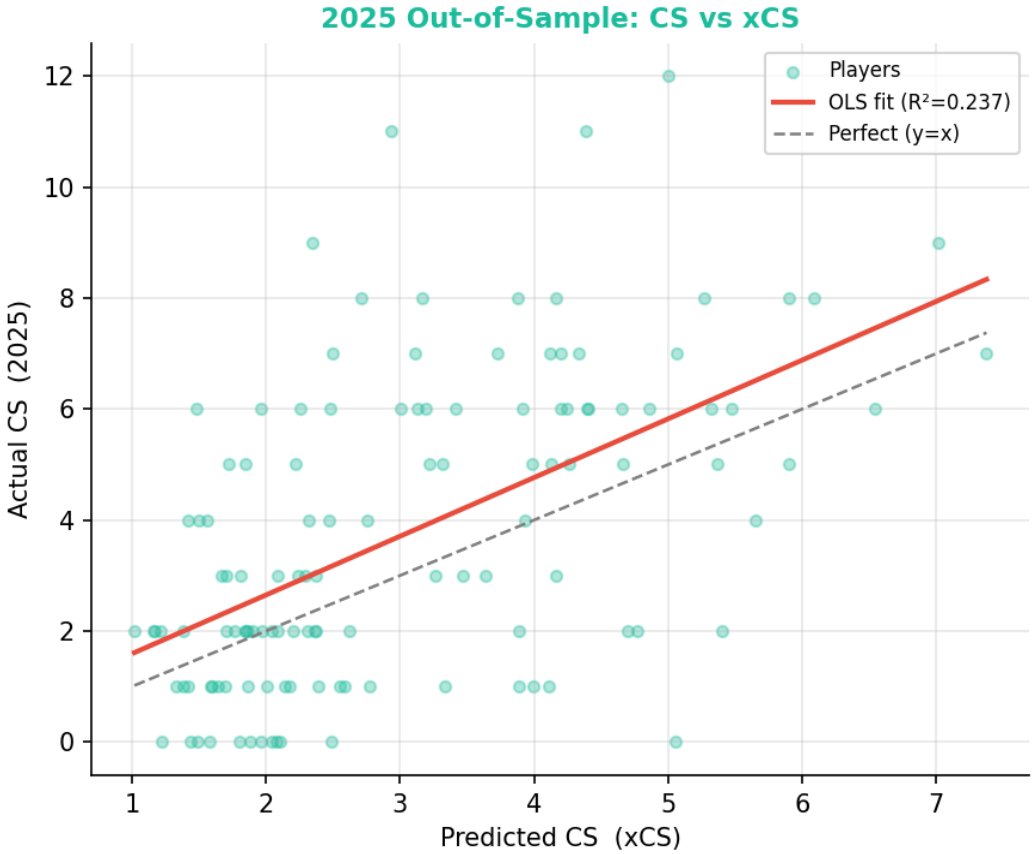
n=122 | R^2=0.5384 | MAE=5.174 | RMSE=7.219 | Mean APE=81.88%

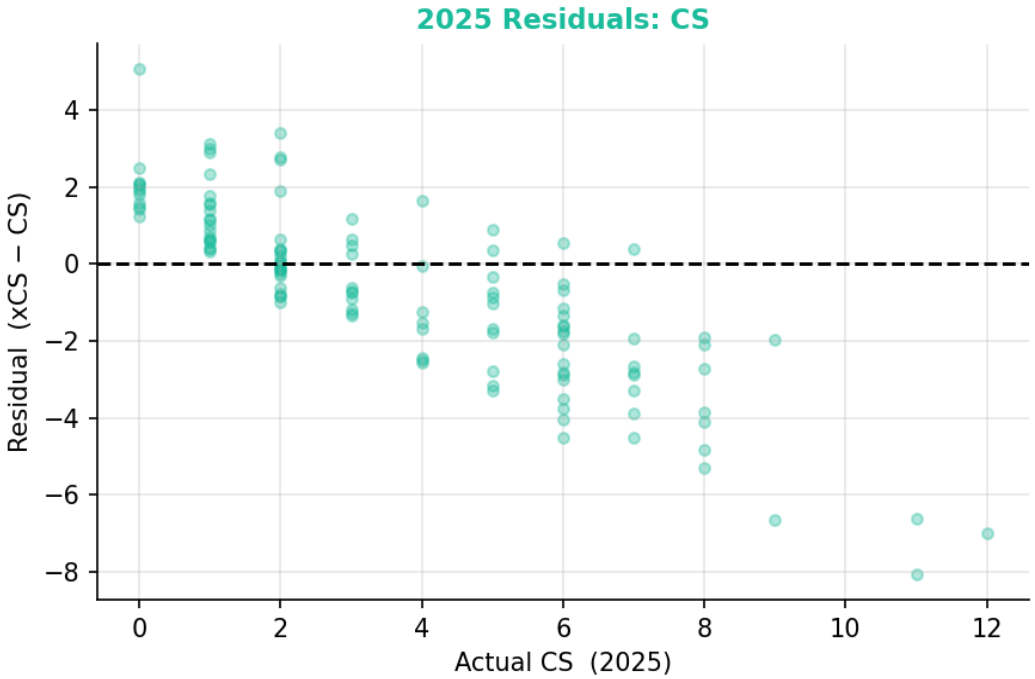




Target: CS

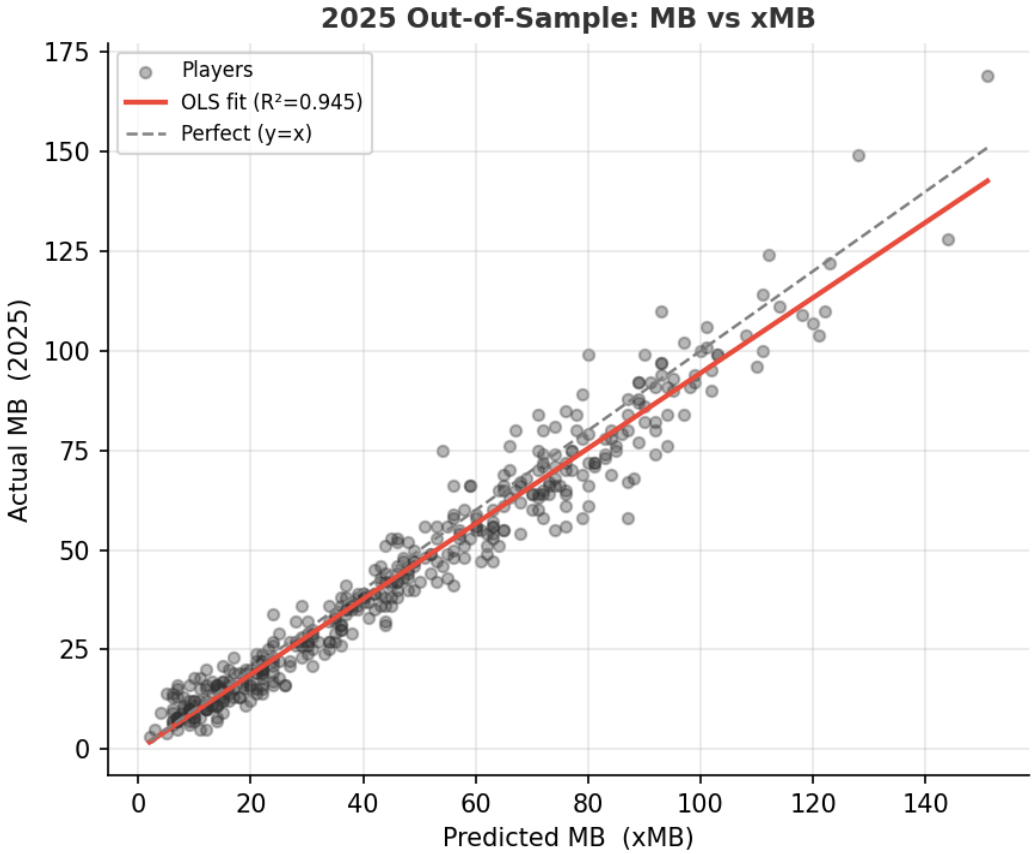
n=122 | R²=0.2369 | MAE=1.854 | RMSE=2.421 | Mean APE=54.75%

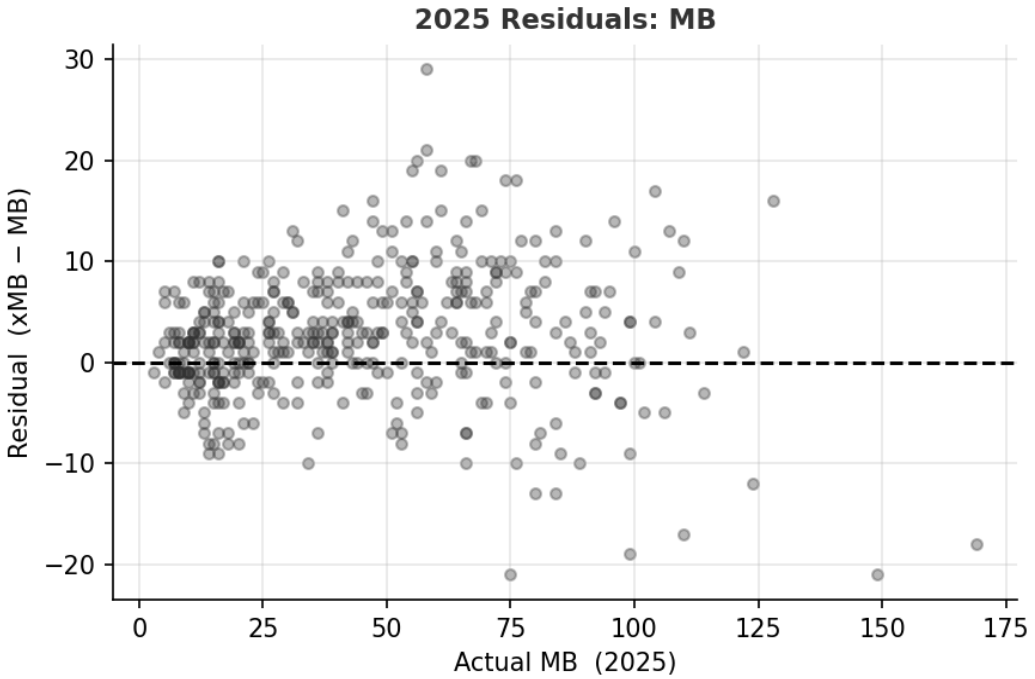




Target: MB

n=425 | R²=0.9449 | MAE=5.238 | RMSE=6.953 | Mean APE=16.12%





Part 3 — 2025 Player-Level Predictions vs Actuals

First 30 rows of the 2025 deployment. Full results saved to **expected_2025_stats.csv**.

Name	Sea son	TB	xTB	BB	xBB	AB	xAB	H	xH	SB	xSB	CS	xCS	MB	xM
Trea Turner	2025.0	269.0	262.04	43.0	41.34	589.0	597.66	179.0	167.14	36.0	37.49	7.0	7.37	99.0	90
Victor Scott II	2025.0	118.0	139.29	42.0	37.45	398.0	388.55	86.0	96.65	34.0	26.6	4.0	5.65	40.0	49
Bobby Witt Jr.	2025.0	312.0	321.11	49.0	46.55	623.0	625.75	184.0	184.84	38.0	35.37	9.0	7.02	111.0	114
Byron Buxton	2025.0	269.0	252.7	41.0	42.3	488.0	486.63	129.0	131.94	24.0	23.09	0.0	5.06	92.0	89
Derek Hill	2025.0	44.0	46.68	9.0	4.83	134.0	146.05	29.0	38.63	7.0	8.23	0.0	2.04	13.0	15
Johan Rojas	2025.0	44.0	47.0	13.0	4.73	152.0	171.47	34.0	38.78	12.0	9.02	2.0	2.2	14.0	13
andon Lockridge	2025.0	40.0	40.88	8.0	0.0	134.0	165.95	31.0	41.86	10.0	8.92	1.0	2.18	12.0	12
Eli White	2025.0	104.0	102.35	11.0	9.77	256.0	255.85	60.0	66.97	10.0	10.64	1.0	2.58	29.0	31
Jake McCarthy	2025.0	71.0	81.3	10.0	14.5	206.0	213.02	42.0	51.19	6.0	10.16	0.0	2.49	18.0	25
Bryce Teodosio	2025.0	42.0	43.02	5.0	0.0	138.0	176.43	28.0	33.37	7.0	8.04	1.0	2.01	10.0	9.
Corbin Carroll	2025.0	305.0	286.65	67.0	54.21	564.0	581.03	146.0	161.18	32.0	30.06	6.0	6.55	106.0	10.
handler Simpson	2025.0	143.0	122.94	20.0	16.28	414.0	380.71	122.0	111.7	44.0	23.76	12.0	5.0	51.0	44
Tyler Fitzgerald	2025.0	71.0	72.07	17.0	9.77	217.0	237.76	47.0	47.83	9.0	8.92	4.0	2.31	19.0	17
Denzel Clarke	2025.0	55.0	37.44	6.0	12.06	148.0	140.02	34.0	28.53	6.0	7.28	0.0	1.89	15.0	11
e Crow-Armstrong	2025.0	284.0	274.06	29.0	38.76	591.0	583.02	146.0	158.1	35.0	22.99	8.0	5.27	82.0	90
Brenton Doyle	2025.0	189.0	220.76	30.0	38.86	502.0	471.34	117.0	125.72	18.0	16.07	2.0	3.89	55.0	74
Edmundo Sosa	2025.0	114.0	92.17	10.0	0.0	243.0	267.76	67.0	70.91	1.0	8.42	1.0	2.14	34.0	24
Michael Helman	2025.0	45.0	58.49	7.0	0.0	99.0	131.18	23.0	12.22	4.0	5.92	0.0	1.58	14.0	6.
Otto Kemp	2025.0	81.0	82.24	12.0	15.05	197.0	192.04	46.0	42.51	2.0	6.82	2.0	1.85	22.0	22
Jonatan Clase	2025.0	30.0	35.84	10.0	14.17	100.0	92.67	21.0	8.18	3.0	6.2	1.0	1.64	9.0	8.
Myles Straw	2025.0	98.0	101.08	19.0	14.12	267.0	273.79	70.0	78.64	12.0	11.32	1.0	2.78	33.0	35
Cam Smith	2025.0	158.0	173.79	43.0	39.44	441.0	463.42	104.0	114.98	8.0	16.04	1.0	4.11	49.0	55
David Hamilton	2025.0	59.0	68.34	13.0	15.47	177.0	171.07	35.0	44.74	22.0	9.07	6.0	2.25	16.0	24
Jacob Young	2025.0	93.0	116.54	27.0	29.34	324.0	319.91	75.0	84.12	15.0	11.81	11.0	2.93	26.0	36
Lane Thomas	2025.0	34.0	38.65	14.0	13.33	125.0	118.44	20.0	6.87	4.0	5.76	1.0	1.59	9.0	6.
Will Benson	2025.0	100.0	115.79	16.0	21.68	230.0	217.08	52.0	51.38	2.0	6.96	2.0	1.85	27.0	35
Tyrone Taylor	2025.0	99.0	107.88	16.0	13.92	310.0	308.15	69.0	76.36	12.0	8.86	2.0	2.38	27.0	32
Oswald Peraza	2025.0	61.0	82.38	17.0	14.09	244.0	245.99	40.0	48.41	9.0	7.9	0.0	2.11	15.0	21
Matt McLain	2025.0	175.0	203.98	55.0	52.17	510.0	515.59	112.0	121.72	18.0	21.6	2.0	5.4	55.0	65

Name	Sea son	TB	xTB	BB	xBB	AB	xAB	H	xH	SB	xSB	CS	xCS	MB	xM
ackson Chourio	2025.0	254.0	244.78	30.0	31.89	549.0	534.9	148.0	149.15	21.0	18.82	7.0	4.33	78.0	79

Part 4 — MB vs xMB: Bayesian Comparison (2025 Data)

The Bayesian comparison now uses **2025 actual MB** vs **xMB predicted by the 2019–2024 model**. This is a true out-of-sample test of whether xMB carries meaningful predictive signal about MB in an unseen season.

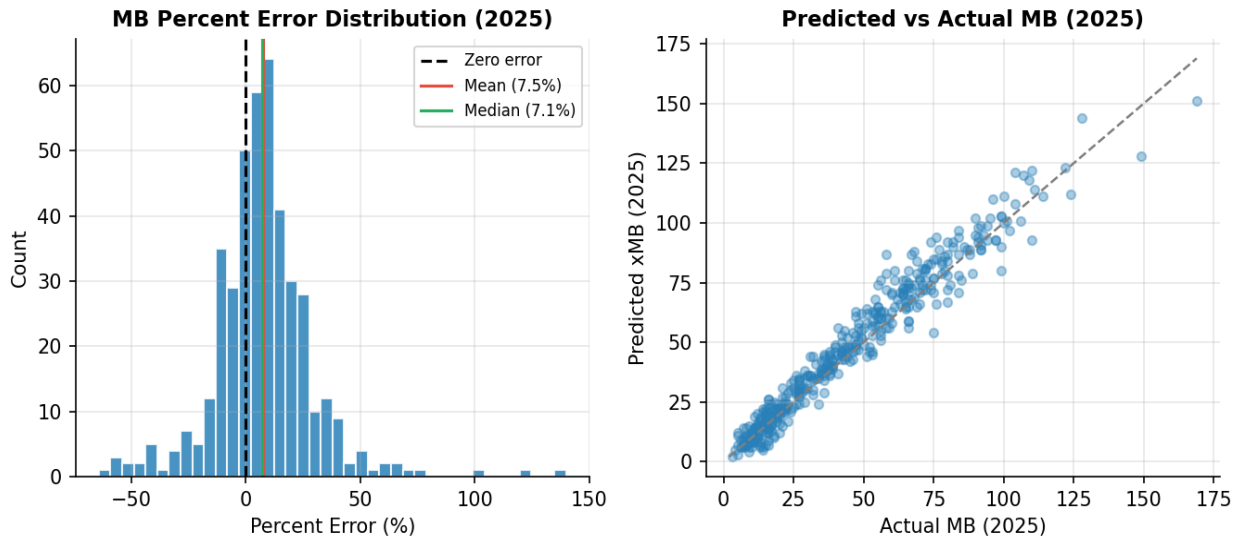
xMB Component Selection

Each of the 6 component stats is evaluated against the R^2 threshold of **0.5**. If a component's best training R^2 meets the threshold, the *expected* (x<stat>) value is used in the xMB formula; otherwise the *actual observed* stat is used.

Component	Training R^2	Threshold (≥0.5)	Used in xMB
H	0.9569	✓ Pass	xH
BB	0.8929	✓ Pass	xBB
CS	0.2863	✗ Fail	CS
TB	0.9541	✓ Pass	xTB
SB	0.4099	✗ Fail	SB
AB	0.9864	✓ Pass	xAB

Components marked ✗ Fail use the real observed stat in the xMB formula.

Metric	Value
Players analysed	425
Mean Percent Error	7.54% (+ = over-predict)
Median Percent Error	7.14%
Mean Abs. Percent Error	16.12%
Median Abs. Percent Error	11.54%
Std Dev of Percent Error	21.83%
Min / Max Percent Error	-64.29% / 140.0%
Players within ±5%	22.4%
Players within ±10%	43.8%
Players within ±20%	74.4%



Left: distribution of MB percent error in 2025. Right: xMB vs MB scatter.

Top 5 Over-Predicted (xMB >> MB)

Name	Season	MB	xMB	% Error
Hunter Feduccia	2025	5.0	12.0	140.0%
Ben Rortvedt	2025	5.0	11.0	120.0%
Cooper Hummel	2025	7.0	14.0	100.0%
Matt Vierling	2025	8.0	14.0	75.0%
Jack Suwinski	2025	11.0	19.0	72.7%

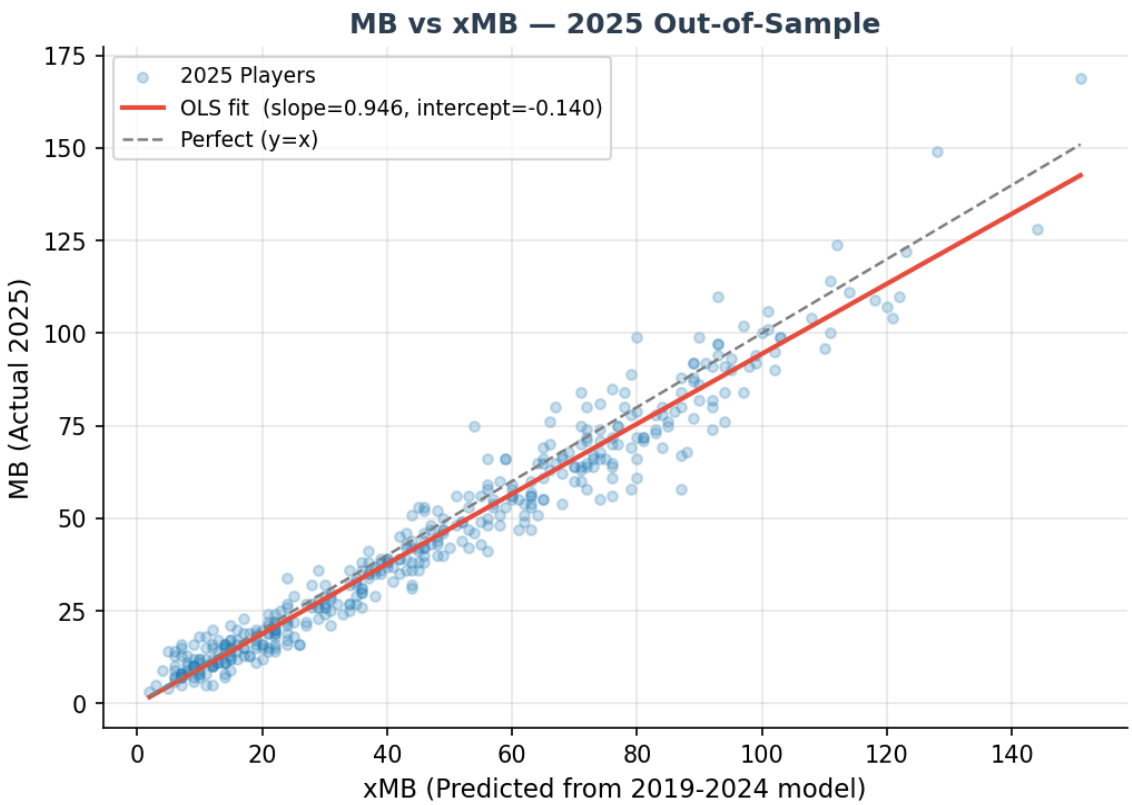
Top 5 Under-Predicted (xMB << MB)

Name	Season	MB	xMB	% Error
Adrian Del Castillo	2025	14.0	5.0	-64.3%
Michael Helman	2025	14.0	6.0	-57.1%
Joey Loperfido	2025	16.0	7.0	-56.2%
Braxton Fulford	2025	9.0	4.0	-55.6%
Marcelo Mayer	2025	13.0	6.0	-53.9%

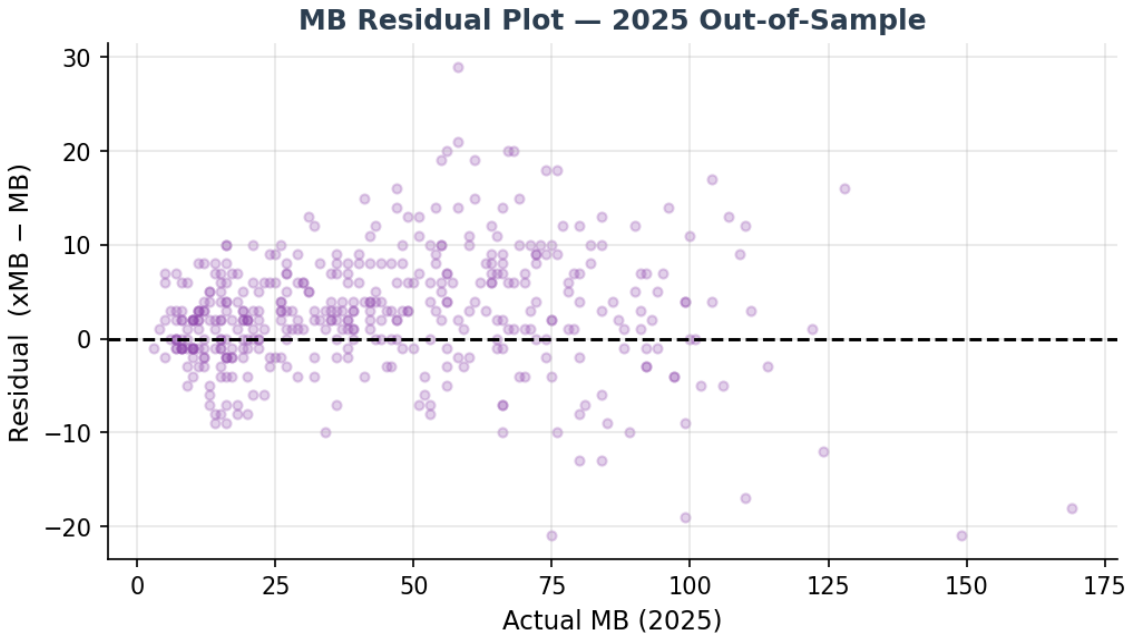
Bayesian Model Comparison (2025)

Statistic	Value
BIC — Model 0 (Null)	2886.52
BIC — Model 1 (xMB predictor)	1561.64
Delta BIC (BIC■ – BIC■)	1324.88
Bayes Factor (BF■■)	4.945195649273331e+287
log■■ Bayes Factor	287.69
Posterior P(Model 1 | 2025 data)	100.0%
OLS Intercept (a)	-0.14
OLS Slope (b)	0.946
Model 1 R^2	0.9564

Verdict (Jeffreys' Scale): Decisive evidence for Model 1



Scatter of 2025 actual MB vs model-predicted xMB. Slope near 1 and intercept near 0 = good calibration.



Residual plot: xMB – MB vs actual MB for 2025.